2025 12th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)
Amity University, Noida, India. Sep 18-19, 2025

2025 12th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO) | 979-8-3315-5421-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICRITO66076.2025.11241350

Leveraging Deep Learning and Explainable AI For Diagnosis of Prostate Cancer

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***Abstract*—This research explores the use of explainable artificial intelligence techniques in diagnosis of prostate cancer. This study uses whole slide images for prostate cancer detection. The study uses various deep learning models such as CNN, ResNet50, VGG19, DenseNet, MobileNet, Xception for image classification. VGG19 outperformed all the other models based on the evaluation metrics. GRAD-CAM, an explainable AI technique was applied to VGG19 model to understand the outcome of the classification. The study's results are encouraging because they showed not only better accuracy but an interpretation of the result using XAI techniques. Following these disclosures, a methodical classification procedure was implemented, dividing the Gleason grades into their respective classes.**

Keywords—Prostate Cancer, XAI, Explainable Artificial Intelligence, Artificial intelligence, Deep Learning, Convolutional Neural Networks, Gleason, Biopsy, Whole slide images

I. Introduction

Prostate cancer ranks the most prevalent cancers and second leading cause of death in men [1]. A major challenge in its diagnosis is that men experience very few symptoms or no symptoms at all [2]. Although benign, it can be detected only in its most advanced stages. The diagnostic methods are prostate biopsies (to perform), Digital Rectal Exam (DRE) and, or systematic Transrectal Ultrasound (TRUS) and Prostate-Specific Antigen (PSA). According to Loeb et al (2019) [3], the limit of over diagnosis and overtreatment of prostate cancer are enormous. Gleason score is usually used for its pathological grading. [4]. The International Society of Urological Pathology (ISUP) recently suggested Grade Groups (GGs) are to improve prognosis prediction. This includes GG1 (non-cancerous), GG3, GG4 and GG5 [5]. However, the development of Whole Slide Imaging (WSI) technology brought about a new era in histopathology. WSIs are the digital scanning of the entire biopsy slide which facilitates detailed analysis and remote diagnostics [6]. Each slide is broken into smaller patches and image features are extracted at multiple scales to capture both fine details and broader contextual information, where imbalance may exist [7]. However, interpreting WSIs is labor intensive and generally requires pathologist's expertise [8]. Owing to improvements in Artificial Intelligence (AI), field of medical diagnostics has seen major developments. From personalized medicine to predictive analytics to image identification, the successful use of AI has touched all areas of healthcare. AI algorithms use MRI scans for analyzing medical images while WSIs for cancers.

Explainable Artificial Intelligence (XAI) is a set of techniques used in AI enabling the results provided by the AI system to be made interpretable to experts. Specifically, it concerns the development of AI models with explanatory feedback capabilities that can be interpreted in a human-understandable way [9]. These explanations will be useful by providing insights into its strengths and weaknesses as well as verifying its reliability.

This study aims to determine whether XAI techniques can be used to improve the transparency and understandability of AI algorithms used for diagnosis of prostate cancer based on WSIs. This paper explores the potential and explainability of AI systems which directly leads to trust and improved patient outcomes in diagnosis of prostate cancer.

Thus, this study aims to enhance transparency and understandability of AI driven diagnosis of prostate cancer using XAI. This work thus employs deep learning models and uses hybrid model (VGG19+ResNet50). This analysis is supported by employing XAI technique known as Grad-CAM.

II. Literature Review

This section present the reviewed literature on prostate cancer diagnosis using various deep learning methods and XAI. There is demand for deep learning models being able to be explainable particularly where quick decision making is needed, like medical image analysis. [10] used an XAI method called Shapley Additive explanations (SHAP) to predict and identify prostate cancer tissue using gene expressions. A study [11] introduced Class Activation Mapping (CAM), an XAI method to differentiate fundus images based on the level of diabetic retinopathy . [12] used an ensemble of Inception-V3, ResNet-152, and Inception-ResNet-V2 for diabetic retinopathy localisation. Similarly, [13] used combination of four CNNs: VGG-16, ResNet-50, Inception-V3, and Inception-ResNet-V2 and used CAMs for detecting acute cerebral hemorrhage. An XAI technique called Grad-CAM

(Gradient-weighted CAM) was introduced by [14], which is a generalization of CAM. The difference between both is that CAM requires global average pooling while Grad-CAM does not and works with all CNN types. [15] [16] utilized Grad-CAM to reveal regions of brain MRI images for identification of a tumor. Similarly, [17] used another XAI technique called Layer-wise Relevance Propagation (LRP) for Alzheimer's disease. Deep SHAP which is a combination of DeepLIFT with Shapley values [18] was employed by [19] to calculate the volumetric breast density from breast MRI data. [20] introduced grid attention and used trainable attention. By incorporating attention gates into a U-Net and a VGG variant, they were able to demonstrate excellent segmentation and localization performance. The regions of the image on which the network concentrated were explained by the attention coefficients. There have been a few studies involving the use of XAI techniques in prostate cancer diagnosis. [21] used CAM to interpret the CNN model. Similarly, [22] utilized CAM for automatic diagnosis and grading of PC on WSIs. [23] employed Local Interpretable Model-agnostic Explanations (LIME) to determine the area of interest for verifying a DL model classification as benign or malignant from ultrasound and MRI images. [24], an interactive XAI model was developed to diagnose prostate cancer. The model accurately identified and categorized clinically significant prostate cancer, boosted non-experts' confidence and reduced their reading time as well as providing both textual and visual explanations based on well-established imaging features.

However, even with the advancement of DL techniques in the healthcare domain, some research gaps persist. One of such is in the context of using WSIs and hybrid models for the analysis in addition to using XAI techniques. not only fills these gaps but also establishes a standard for other studies that seek to use XAI and DL to address healthcare issues.

III. Methodology

This study utilized an experimental approach. Fig. 1 depicts the proposed workflow.

Data was collected from a database known as SICAPv2 [25]. The is currently the largest publicly available set of whole slide image prostate H&E biopsies with pixel level annotation. A total of 4417 non-cancerous patches are present in the data, 1635 are labelled as GG3, 3622 as GG4 and 665 as GG5. In this work, WSI were analyzed using image processing techniques to enhance the images and retrieve relevant information. All WSIs were stored in JPEG format and the data was split into training (80%) and validation sets (20%) with equal ratios of images from each class. To fit the model, the JPEG images were resized to 224 x 224 x 3 or 299 x 299 x 3 depending on the algorithm employed. Normalization was performed by dividing each pixel value by 255. Regardless of the colors that were actually present in the images, this was done for every channel.

Different forms of image augmentation were employed including shifting, flipping and zooming the images. The model may overfit if image augmentation is not employed. By intentionally increasing the training dataset and adding changes to the images, image augmentation can assist avoid overfitting, enabling the model to understand robust features. The trained models are fine-tuned to better fit the models. Class weighting is used for data balancing. Deep learning and XAI techniques were then applied for detection of prostate cancer.

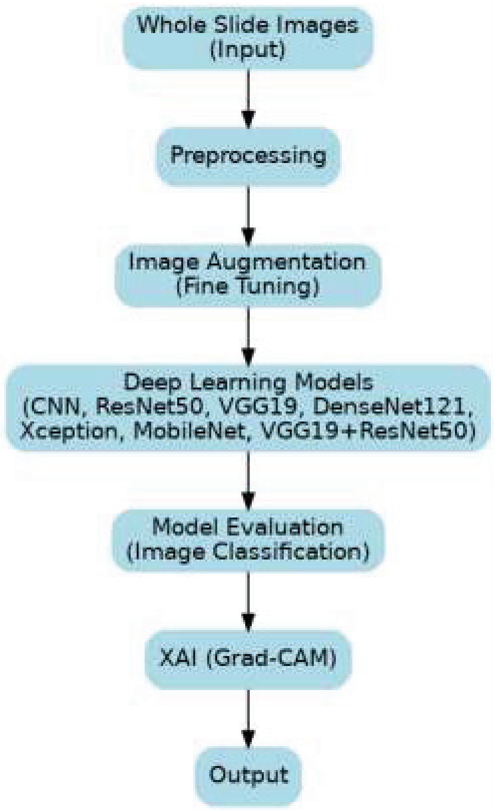

Fig. 1. Proposed workflow

IV. Baseline Models

Different baseline models were used in comparison with proposed models.

A. CNN

A CNN is an artificial neural network designed to process input that has a grid-like structure, like an image. It is powerful in understanding spatial hierarchies and hence it is appropriate for analyzing images. CNNs consist of several layers viz. convolutional, pooling and fully connected layers. A convolutional layer creates feature maps by sliding filters over the input data to highlight regions of interest. The pooling layers down samples the data, reducing its dimension and strengthening the network's resistance to even small translations in the input data. Ultimately, the fully connected layers map the characteristics that the convolutional layers collected to the desired output [26], [27], [28].

B. MobileNet

This model can be used in contexts with limited resources, such as mobile devices. To accomplish this efficiency, it combines pointwise and depth wise convolutions with a brand new activation function known as swish. It is faster than the larger versions. It accepts an input size of 224x224 [29].

C. Xception

Extreme version of Inception is a CNN model developed by Google. It is pre-trained on ImageNet data and consists of 71 layers [30].

D. DenseNet121

Densely Connected Convolutional Network (DenseNet) is a feed-forward CNN that connects each layer with every other layer in the architecture. This design reduces of the number of parameters and enhances the gradient flow, enabling the network to learn more efficiently through effective feature reuse [31].

E. GRAD-CAM

This research uses XAI, the Gradient-weighted Class Activation Mapping (Grad-CAM) technique to determine areas of an input image that are crucial for deciding which class the network will predict. It is a method that provides interpretability without sacrificing accuracy while still maintaining the architecture of deep learning models [32].

V. PROPOSED MODELS FOR COMPARISON

The following models are proposed to be used for detection of prostate cancer along with a hybrid model combining ResNet50 and VGG19.

A. ResNet50

ResNet50, is a CNN containing 50 layers with 48 convolutional layers, 1 max pooling layer and 1 average pooling layer. It builds networks by stacking residual blocks [33].

B. VGG19

VGG19 is a CNN with 19 layers and is known for its efficacy and efficiency in image classification tasks. It consists of 16 convolutional layers and 3 fully connected layers. The convolutional layers of the network are built using a mix of 3 x 3 and 1 x 1 filters, and the network is trained on ImageNet dataset, which has over one million annotated images [33].

C. VGG19-ResNET50

The hybrid model is composed of both VGG19 and ResNet50 algorithms. Both models are loaded without the top layers and maintain the same input and output shape. The output of the models is then concatenated to create the hybrid model. The output layer consists of a dense layer with softmax activation function. In this research, one of the aims is to check the performance of this hybrid approach and compare the results with that on a single model approach to see which performs better.

VI. RESULTS AND DISCUSSION

Fig. 2 shows the performance of different models based on the accuracy. With an accuracy of 0.78, VGG19 and Xception models stand out. This accuracy is essential to reduce the possibility of misdiagnosing patients wrongly, which can be detrimental in real world scenarios. With an accuracy of 0.41, the CNN model raises concerns about its dependability for this task. Since this is a multi-classification problem, a low accuracy may lead to an inability to correctly diagnose the Gleason grades.

The precision of prostate cancer diagnosis is particularly important because false positives may provide needless treatments or divert attention from actual patients with prostate cancer. The VGG19 and Xception models have a precision score of 0.75, indicating that there is a high probability of precision when this model classifies a Gleason grade or diagnose prostate cancer. The right allocation of resources and attention depends on this level of precision.

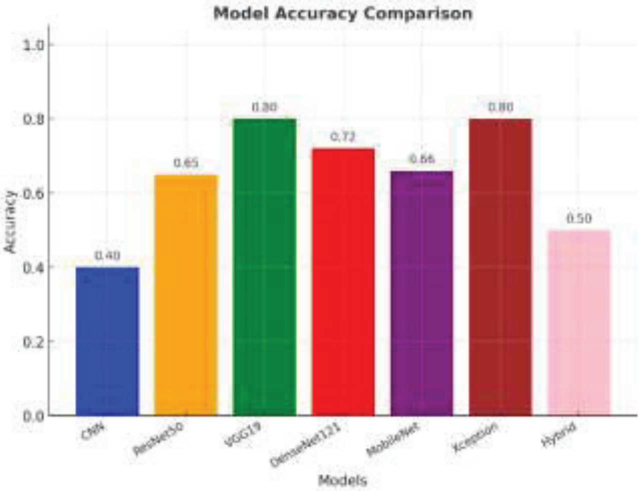

Fig. 2. Model comparison based on accuracy

The recall is crucial because it could have major ethical and safety implications if one fails to recognize a prostate cancer biopsy correctly. With a recall score of 0.77, VGG19 ensures to a degree that the model can accurately identify the majority cases pertaining to prostate cancer. Reduced recall rates—like those of the CNN and Hybrid algorithms may lead to missed chances to offer assistance or treatment to those who are living with prostate cancer.

The F1-Score is a composite metric balancing precision and recall thus offering comprehensive assessment of model performance. For the healthcare sector, it is a reliable statistics for assessment where the cost of false positives and false negatives can be high. The F1-Score of 0.75 for VGG19 suggests a reasonable capacity to recognize and accurately categorize stages of prostate cancer.

ROC AUC Score of the various models shows overall performance of the model. It tells us how the model is performing, either it is guessing or learning. The VGG19 model has the highest ROC AUC score of 0.85, which shows that the model can reliably identify and differentiate between the different stages of prostate cancer. This ensures that patients receive accurate diagnosis and appropriate treatment plans. The low ROC AUC score in the CNN model shows that the model is close to random guessing and should not be used in the real world.

With scores averaging 0.75, VGG19 constantly ranks highest across all parameters, demonstrating the best possible trade-off between high accuracy, precision and recall. On the other hand, the CNN model has a low accuracy (0.41) and F1-score (0.24) (Table 1). These differences in algorithm performance highlight the need for a multi-metric assessment methodology, particularly for tasks involving safety and ethical considerations. The VGG19 model's effectiveness as a prostate cancer diagnostic tool is reinforced by the comparison analysis conducted across these measures, which shows that the model can minimize errors while optimizing correct classifications of the Gleason grades.

TABLE I. EVALUATION METRICS FOR DIFFERENT MODELS

Models	Accuracy	Precision	Recall	F1-Score	ROC-AUC Score
CNN	0.41	0.20	0.29	0.24	0.53
ResNet50	0.67	0.65	0.66	0.63	0.78
VGG19	**0.78**	**0.75**	0.77	0.75	0.85
DenseNet	0.73	0.71	0.75	0.71	0.83
MobileNet	0.66	0.66	0.68	0.64	0.78
Xception	0.78	0.75	0.75	0.74	0.84
Hybrid (VGG19+ResNet50)	0.50	0.53	0.49	0.43	0.66

Since the VGG19 model has been confirmed to be the best performing model, the Grad-CAM was applied for interpretation of the model's decision. In the context of medical images, it is vital to indicate the area of the image that was used for analysis by the model to come to its conclusion. This was done at random for some Gleason grade class to properly highlight areas of the image that the model extracted to arrive at its results. Figs 3-5 show the image segments used by the model for classification of the image as non-cancerous, G4 and G5 respectively.

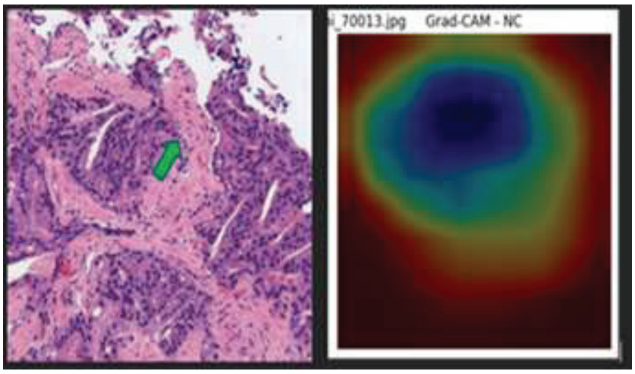

Fig. 3. GRAD-CAM for NC class

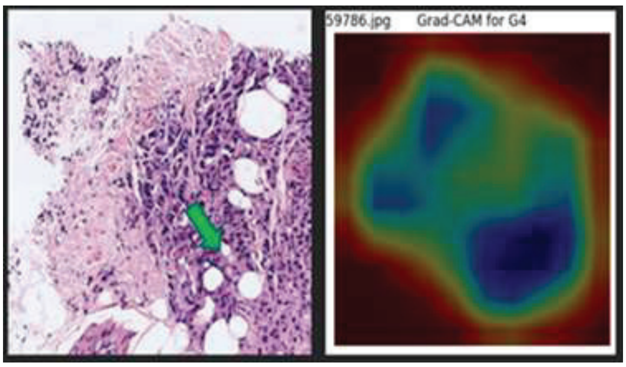

Fig. 4. GRAD-CAM for G4 class

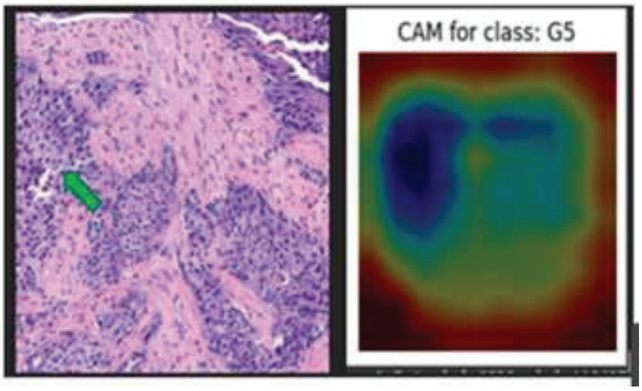

Fig. 5. GRAD-CAM for G5 class

VII. CONCLUSION

The research used deep learning and XAI techniques to develop a system for diagnosis of prostate cancer along with providing justification for the model's decision. Although models are able to correctly diagnose prostate cancer, it will still require advice of expert doctor. The system has not been tested in clinical settings, and so future studies could develop system with validation in clinical settings. Future studies could explore other XAI techniques to get different insights to how the models arrive at their decision.

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